Hardware Setup:

I started with the basic breadboard setup that was covered in class.

Button:

The button is read on GPIO 15, read with Pin.IN, PULL_DOWN, button.value() == 1 means that the button has been pressed.

Two wires connect the button circuit

- Row 61i ↠ negative (ground) rail at row 57
- Row 63c ↠ row b20, which lines up with the Pico's GPIO 15 leg

LED:

We use one two-legged LED as the flashing light when the alarm goes off. Located at GP17, long leg 38d, short leg 40d. The LED has its own resistor and wire. The resistor has a leg that is placed next to the short leg of the LED. The other leg is connected to the negative rail. One leg of the wire sits next to the LED's front leg, and the other leg connects to its corresponding Pico GPIO leg.

- Resistor: 40a ↠ negative rail (row 37)
- Wire to Pico: 38 e↠ row 19j (Pico GP17)

Buzzer:

The two-legged buzzer plays an alarm for the user. It is located at i/20 (GP 16), with its second leg at h22. It has a resistor at i22 connected to the negative rail.

Rotary Potentiometer:

The rotary potentiometer is used to control the buzzer volume. It is connected to the Pico's GPIO 27 and read using the ADC.

- Potentiometer legs: f42, f43, f44
- White Wire: i43 ↠ j11
- Green Wire: j42 ↠ j10

Motion Sensor:

I originally had the motion sensor as part of the project, but it wasn't working and eventually caused my other features to not work as well. Due to this issue, the final working version does not include the motion-based snooze feature.